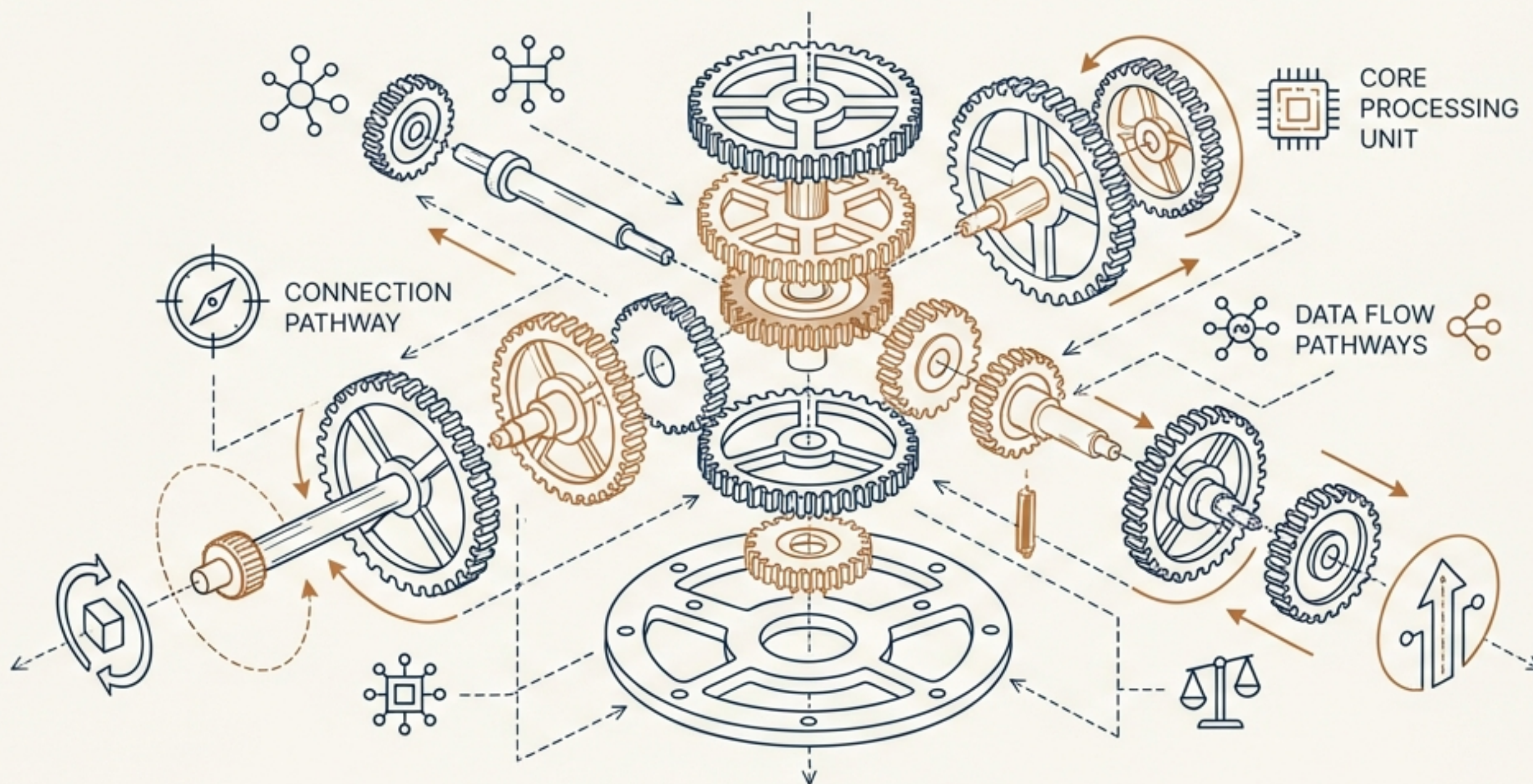


Phase 0: Blueprint for a Digital Society

Architecting the Foundational Infrastructure for a Multi-Agent Collective

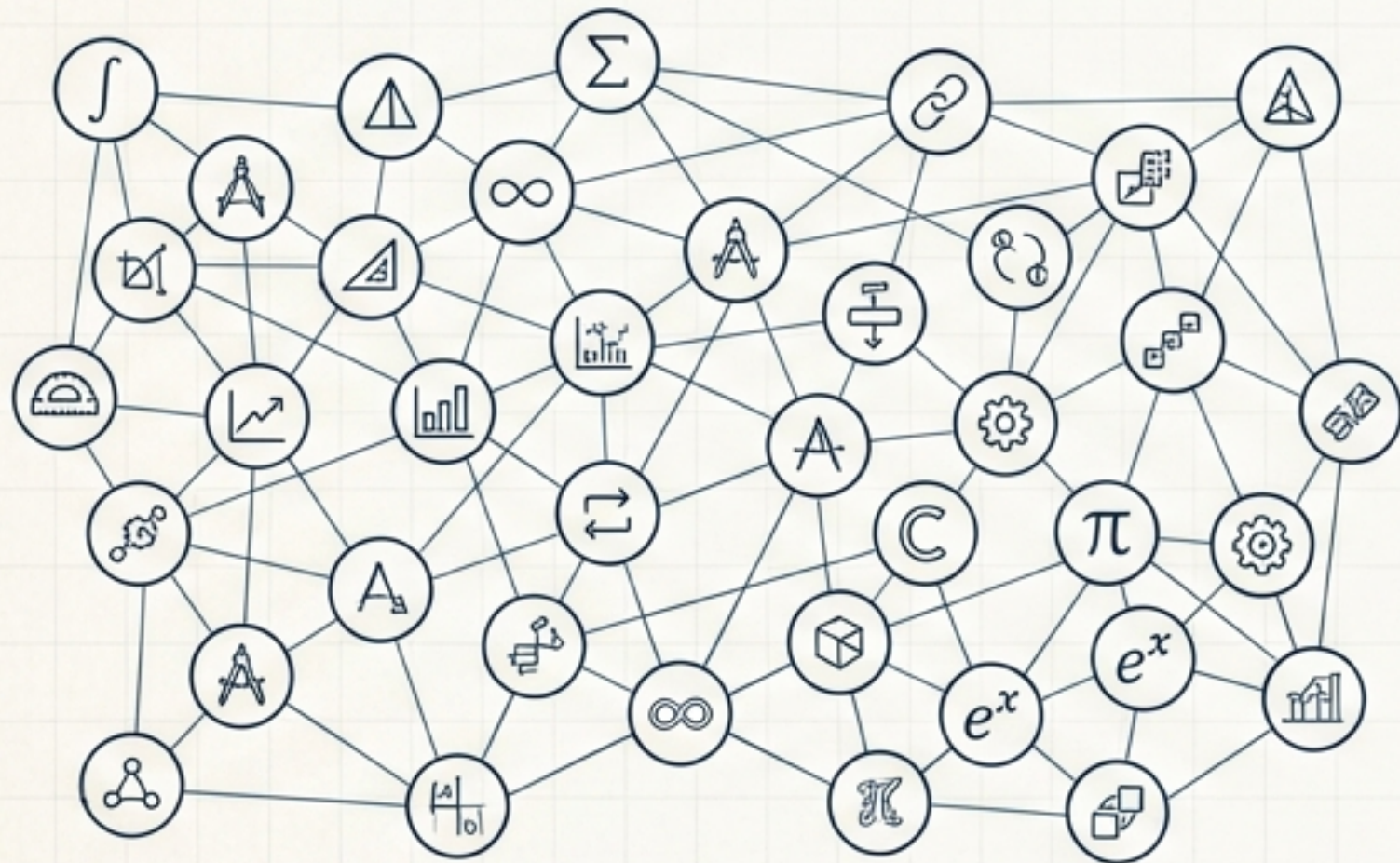


Genesis V2

The Goal: A Society of Specialists Confronts Physical Law

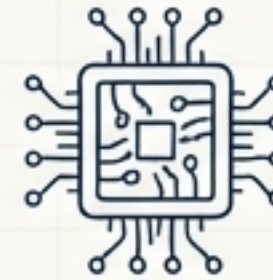
A Thriving Collective

To build a system of **40-50 specialized agents** collaborating to solve complex mathematical problems.

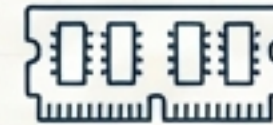


The Hardware Foundation

This society must exist on a specific physical machine.



CPU: AMD Ryzen 7 8700F



System RAM: 32 GB



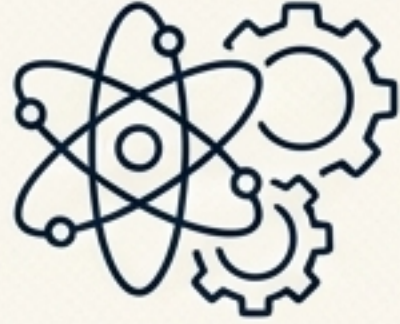
GPU: NVIDIA GeForce RTX 4060



Dedicated VRAM: 8 GB

The Phase 0 Mandate: Build the City Before the Citizens Arrive

Phase 0 is not about writing problem-solving code. It is about codifying the fundamental rules of the system's reality. We are **architecting**:



The Laws of Physics

Hardware-aware protocols that respect our resource limits.



The Official Language & Legal Code

How agents communicate unambiguously and act with intent.



The Civic Infrastructure

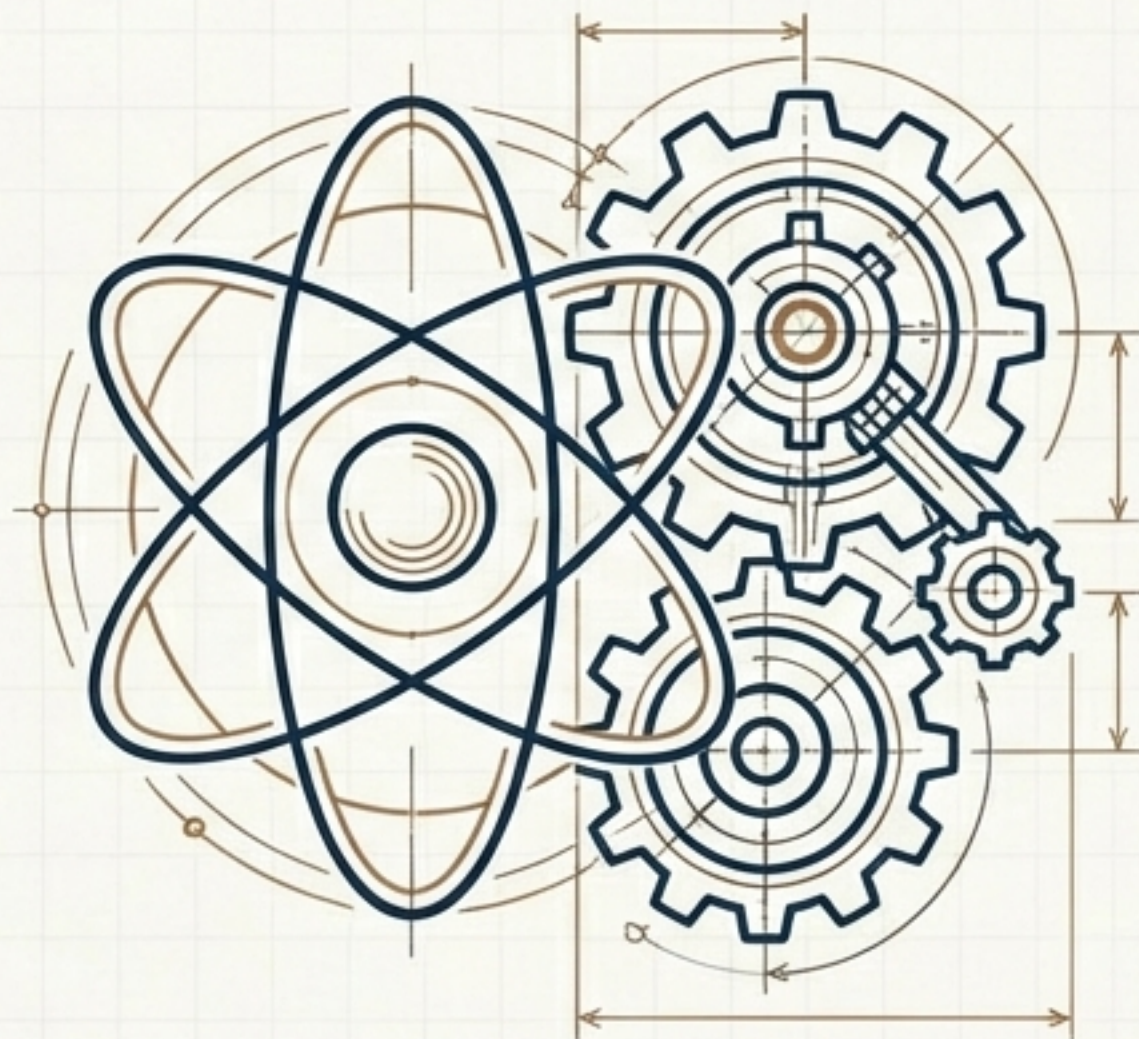
'Silent' agents that manage other agents.



The Public Memory

How knowledge is shared, stored, and retrieved.

Part 1: The Laws of Physics



Calibrating to Hardware Reality

The most fundamental law of our digital society is **resource scarcity**. Our **architecture must be ruthlessly efficient** from the start, addressing the primary bottlenecks of VRAM and context window limits.

Enforcing the 'One-Model-At-A-Time' Rule

Problem

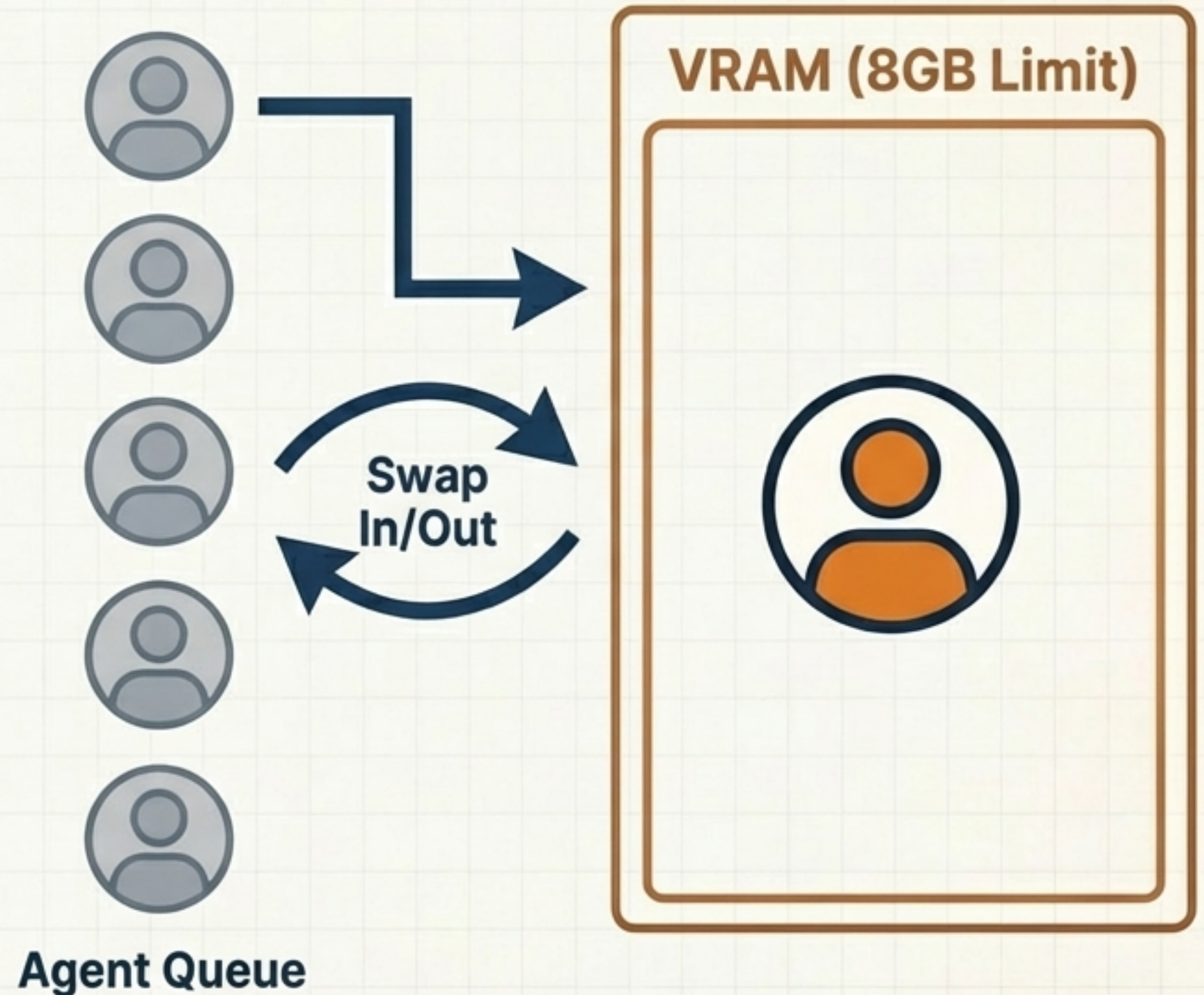
A single 7B parameter LLM requires ~5-6GB of VRAM when quantized. Our 8GB RTX 4060 can barely hold *one* active reasoning agent. Parallel cognitive agents are physically impossible.

Phase 0 Mandate

Implement a **Stateful Serialization Protocol**.

Mechanism: The **Agent Management System (AMS)** acts as a gatekeeper, enforcing a strict rule: only one cognitive agent can occupy VRAM at any given moment.

Analogy: Cognitive can mannteated mechaniate. A high-security lab with only one active workstation. Agents must be swapped in and out.



Part 2: The Civic Infrastructure



The "Silent" Agents of Governance

Before any 'mathematician' agents are instantiated, we must build the **'bureaucrat' agents who manage the city itself. These agents are computationally active but produce no mathematical output.**

The Triumvirate of System Management



Agent Management System (AMS)

Role: The 'City Hall' or 'God Agent'

Type: Simple Reflex Agent

Function: Manages agent lifecycles (create, kill, suspend).

Critically, it monitors it monitors VRAM/RAM usage to enforce the one-model rule.

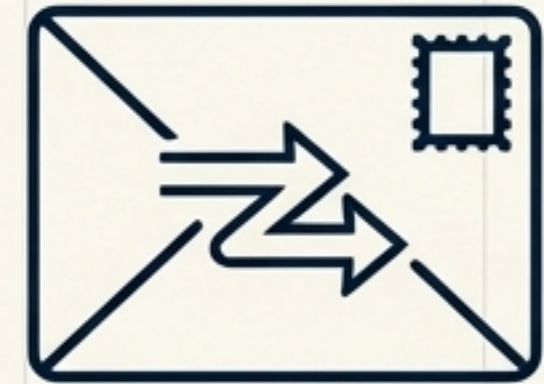


Directory Facilitator (DF)

Role: The 'Yellow Pages' or 'Service Registry'

Type: Model-Based Reflex Agent

Function: Provides a dynamic lookup for agent services (e.g., 'service: integration'). Agents register their capabilities here.



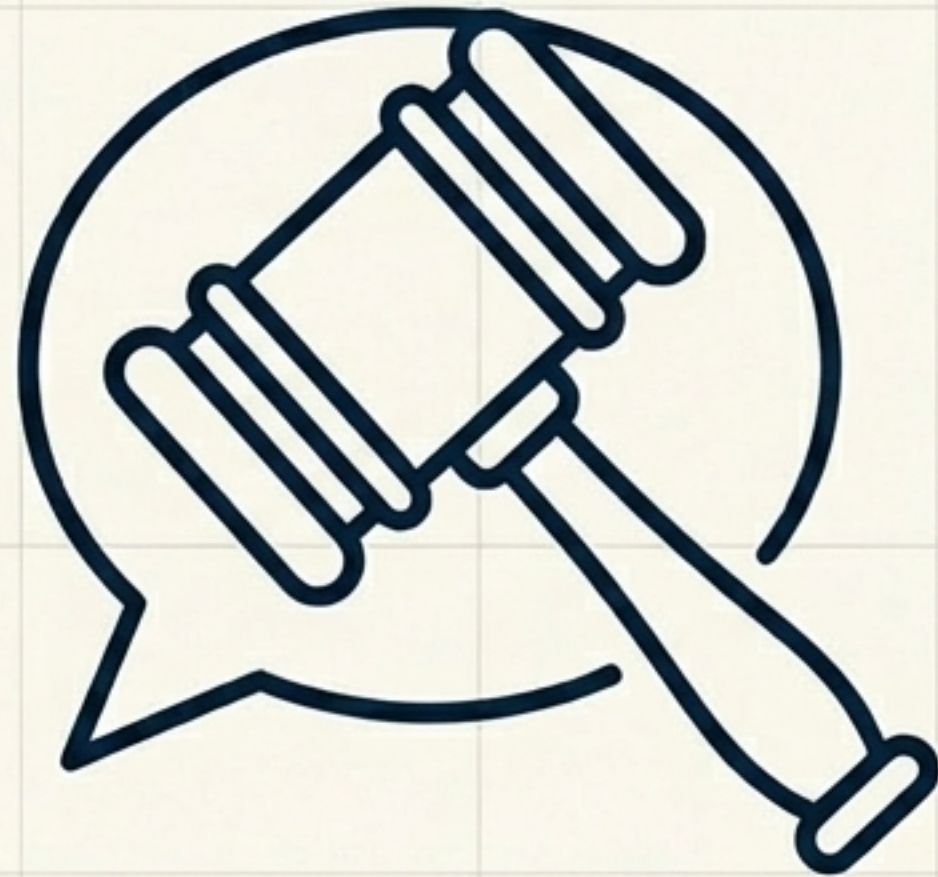
Agent Communication Channel (ACC)

Role: The 'Postmaster'

Type: Transport/Routing Agent

Function: Guarantees routing of FIPA-ACL messages. Queues messages for agents that are currently inactive or swapped out of VRAM.

Part 3: The Official Language & Legal Code



Architecting Unambiguous Communication

To prevent a 'Tower of Babel' scenario, agent communication cannot be based on ambiguous natural language. We must establish a **formal language** and a **legal framework** for interaction.

The Lingua Franca: Speaking Pure Mathematics with OMDoc

Problem

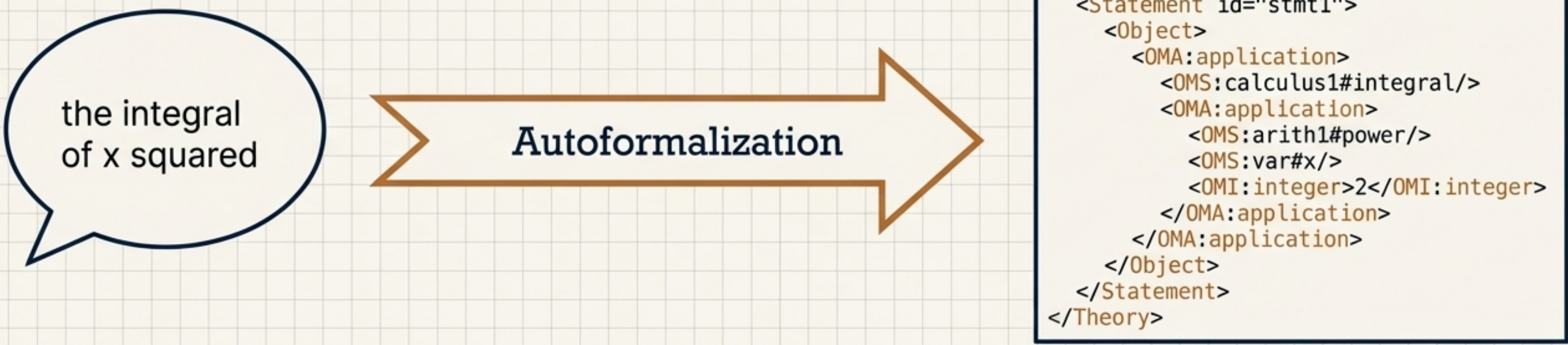
Raw text like '\$x^2\$' is a string of characters; it lacks semantic meaning. It conveys *presentation*, not *content*.

Phase 0 Mandate

All inter-agent communication must use the **OMDoc** (**Open Mathematical Documents**) standard.

Why OMDoc?

It encodes mathematical *meaning*. A “derivative” is understood as a formal operator.



the integral
of x squared

Autoformalization

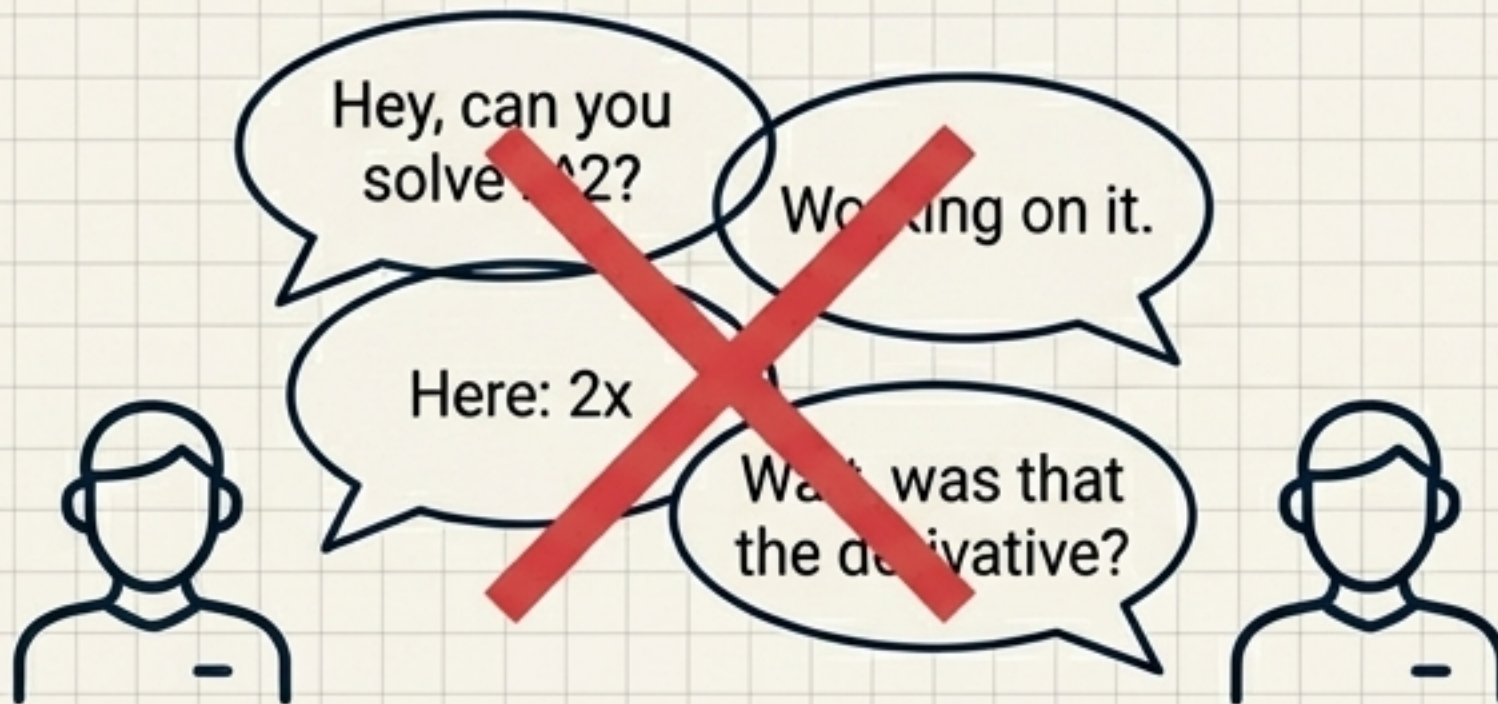
```
<Theory name="Calculus">
  <Statement id="stmt1">
    <Object>
      <OMA:application>
        <OMS:calculus1#integral/>
        <OMA:application>
          <OMS:arith1#power/>
          <OMS:var#x/>
          <OMI:integer>2</OMI:integer>
        </OMA:application>
      </OMA:application>
    </Object>
  </Statement>
</Theory>
```

The Rule of Law: Acting with Explicit Intent via FIPA-ACL

Problem

A message isn't enough. An agent must know the *intent* behind it. Is it a request? An order? A confirmation?

Ambiguous Chat



Phase 0 Mandate

Implement the **FIPA-ACL (Agent Communication Language)** protocol, which is based on **Speech Act Theory**.

Key two mechanisms:

- **The 'Performative' Field:** Makes the sender's intent explicit and binding (e.g., `REQUEST`, `INFORM`, `REFUSE`). A message is an *action*.
- **Mandatory Conversation IDs:** Allows agents to track specific problem-solving threads without passing redundant conversational history, conserving the context window.

Structured FIPA-ACL Message



```
:performative    REQUEST
:sender          Agent_Orchestrator
:receiver        Agent_Calculus_1
:content         {OMDoc Object for "d/dx(x^2)"}
:language        OMDoc
:conversation-id prob_7b3c_thread_1
```

Part 4: The Public Memory



Building the Collective Consciousness

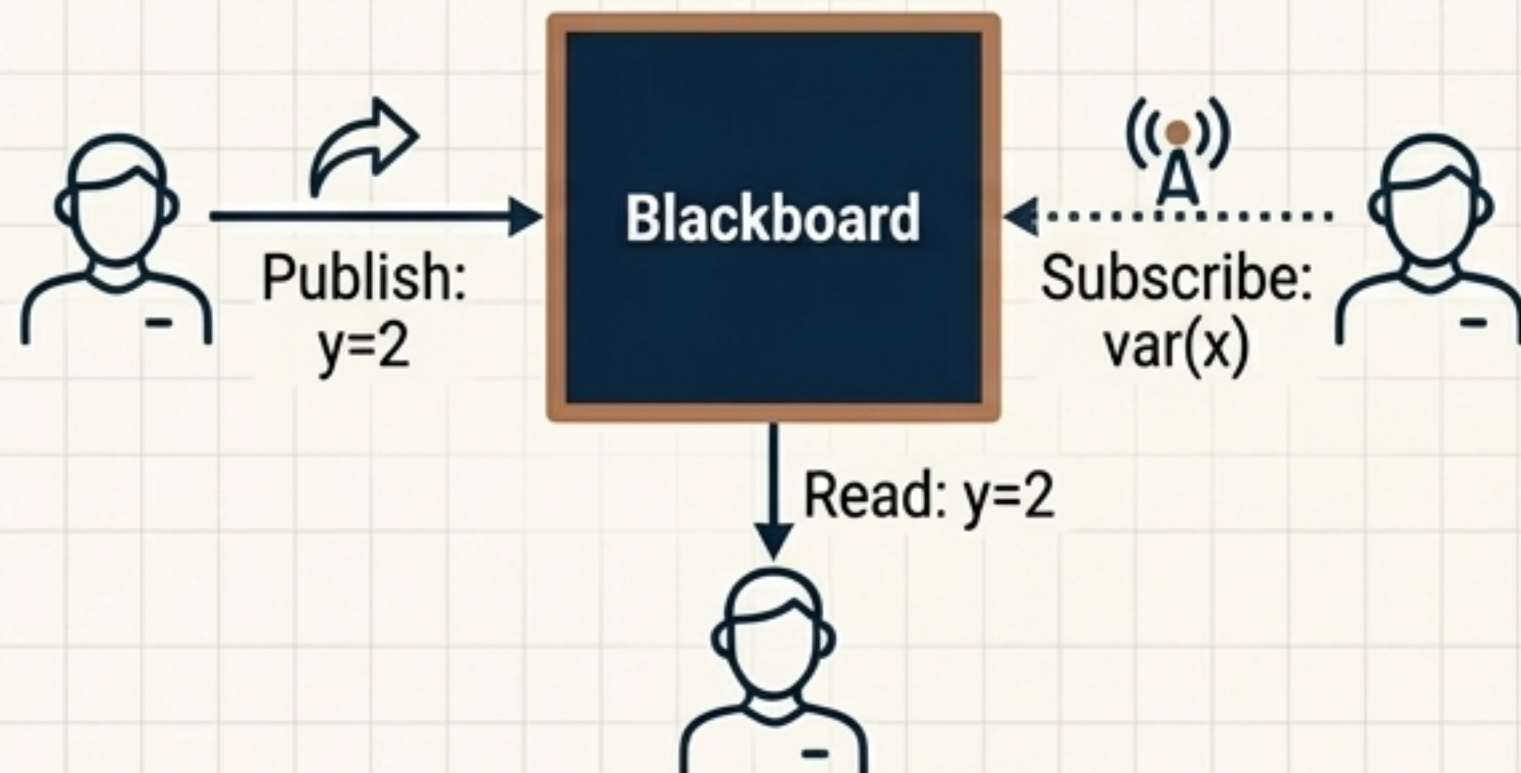
Agents cannot be stateless entities operating in a vacuum. They must contribute to and draw from a shared pool of knowledge, successes, and failures.

A Two-Part Memory: The Workspace and The Archive

The Blackboard: The “Shared Workspace”

Function: A central location for agents to post partial results and lemmas.

Mechanism: A **Publish-Subscribe** system. A Geometry agent can “subscribe” to updates on variable x and be notified when an Algebra agent solves for it, enabling emergent collaboration.

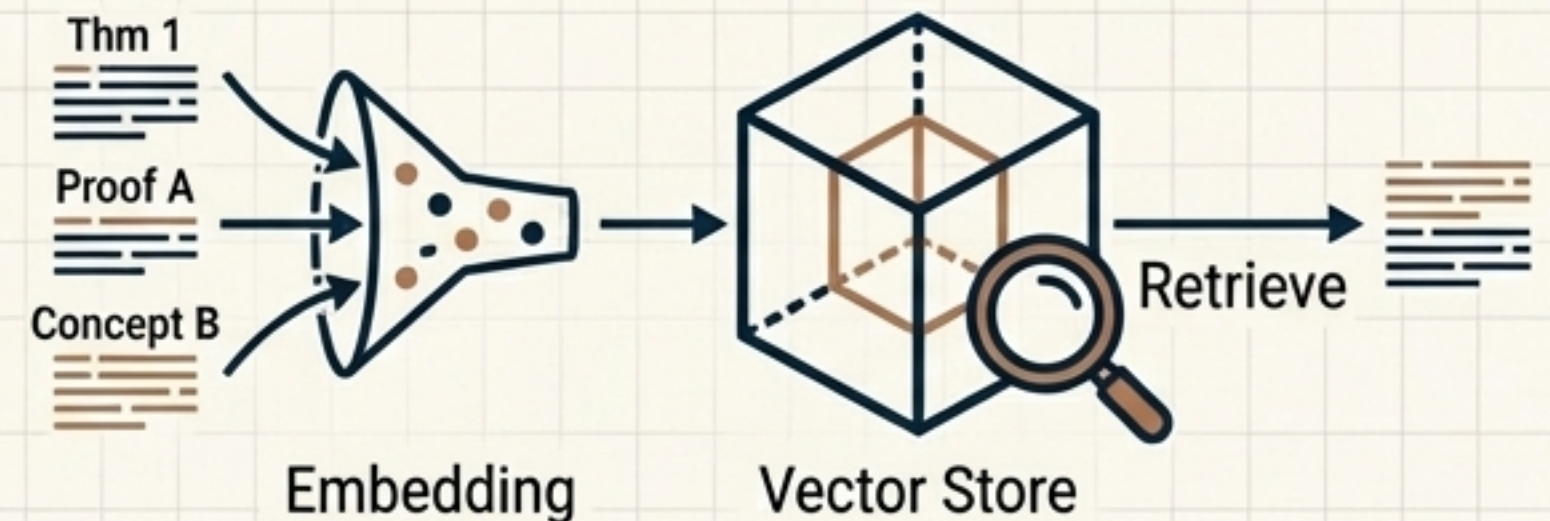


The Vector Database: The “Long-Term Archive”

Function: The prerequisite infrastructure for **Retrieval-Augmented Generation (RAG)**.

Mechanism: Indexes mathematical “thought traces” and theorems, allowing the system to recognize recurring problem patterns over time.

Implementation Note: A lightweight store (e.g., **Milvus**, **Chroma**) is chosen to conserve system RAM.

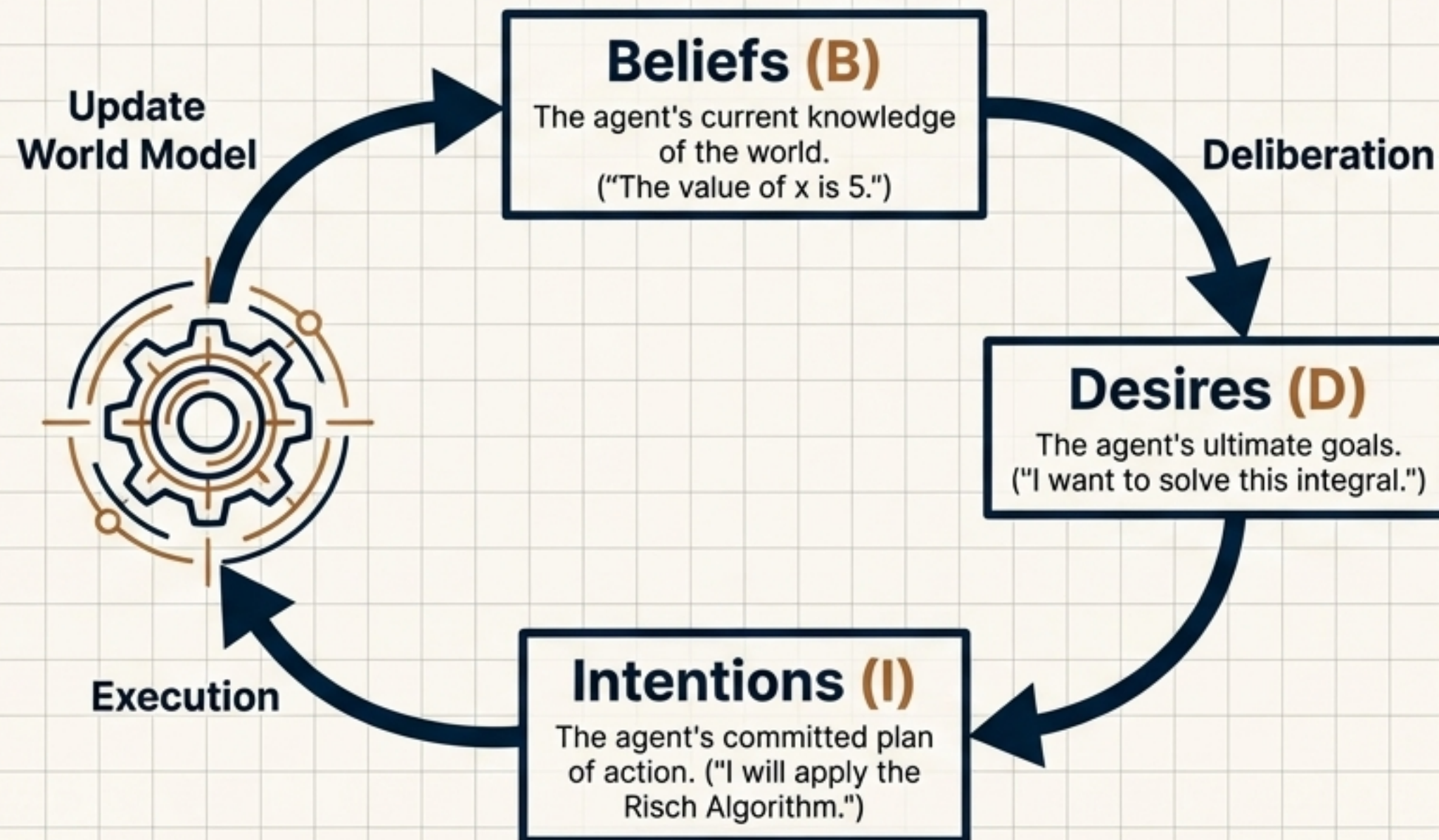


The Cognitive Blueprint for Every Citizen: The BDI Model

We must define the internal “operating system” for every future cognitive agent. They must be deliberative entities, not simple scripts.”

Phase 0 Mandate: All cognitive agents will inherit the **Belief-Desire-Intention (BDI) control loop**.

This architecture separates deliberation (choosing a plan) from execution (doing the math).



Phase 0 Complete:

Computationally Alive, Mathematically Inert

At the conclusion of Phase 0, the system is fully operational but **has not yet solved a single problem**. We have built the city, but the streets are empty.

- ✓ **The Office Building is Built:** Infrastructure is running.
- ✓ **The City Manager is Hired:** The AMS is monitoring resources.
- ✓ **The Laws are Written:** FIPA and OMDoc govern all future interaction.
- ✓ **The Library is Open:** The Blackboard and Vector DB are ready.



The Stage is Set for Phase 1

With this immutable bedrock in place, we can now safely begin **instantiating the first cognitive agents**—the cognitive agents—the mathematicians who will inhabit our city.

Coming in Phase 1:

- ⚙️ The Tier 1 Orchestrator
- 📄 The Autoformalization Pipeline
- 👤 The First Specialist Agents

